Lecture 16: Finite Galois Theory — The Fundamental Theorem

MAT205 Abstract Algebra II

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Part I: The Correspondence (the Goal)

Setup: the Galois Group and the Two Maps

Recall. E/F finite is *Galois* if normal and separable, equivalently $|\mathrm{Aut}(E/F)| = [E : F]$. Its *Galois group* is $G := \mathrm{Gal}(E/F) := \mathrm{Aut}(E/F)$, with $|G| = [E : F]$.

Two maps, between intermediate fields $F \subseteq K \subseteq E$ and subgroups $H \leq G$:

- field $\to$ group: $K \mapsto \mathrm{Gal}(E/K) = \{\sigma \in G : \sigma|_K = \mathrm{id}\}$;
- group $\to$ field: $H \mapsto E^H = \{x \in E : \sigma x = x \ \forall \sigma \in H\}$, the *fixed field*.

Both **reverse inclusions**: $K_1 \subseteq K_2 \Rightarrow \mathrm{Gal}(E/K_2) \leq \mathrm{Gal}(E/K_1)$, and $H_1 \leq H_2 \Rightarrow E^{H_2} \subseteq E^{H_1}$.

Fixed Fields

Definition. For $H \leq \mathrm{Aut}(E)$, the *fixed field* is
$E^H = \{x \in E : \sigma(x) = x$ for all $\sigma \in H\}$ — a subfield of E.

For $H = G = \mathrm{Gal}(E/F)$ always $F \subseteq E^G \subseteq E$, and E/F is Galois **iff** $E^G = F$ (only F is fixed by every F-automorphism). [This is the $K = F$ case of the correspondence; it falls out of Artin's theorem, proved in Part II.]

Example. $E = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, $G = \{\mathrm{id}, a, b, c\} \cong V_4$ with $a : \sqrt{2} \mapsto \sqrt{2},\ \sqrt{3} \mapsto -\sqrt{3}$.
Then $E^{\langle a \rangle} = \mathbb{Q}(\sqrt{2})$ and $E^G = \mathbb{Q}$.

The Fundamental Theorem (Statement)

Theorem (Fundamental Theorem of Galois Theory). For finite Galois E/F with $G = \mathrm{Gal}(E/F)$, the maps

$$K \longmapsto \mathrm{Gal}(E/K), \qquad H \longmapsto E^H$$

are mutually inverse, inclusion-reversing bijections between intermediate fields and subgroups of G, with

$$[E:K] = |H|, \qquad [K:F] = [G:H] \qquad (H = \mathrm{Gal}(E/K)).$$

Dictionary. $E \leftrightarrow \{1\}$, $F \leftrightarrow G$; bigger field $\leftrightarrow$ smaller group; degree over $F \leftrightarrow$ index. (Made concrete on the worked examples.)

Two further parts, proved once we have Artin's theorem: the bijection is a *lattice anti-isomorphism*, and K/F is itself Galois exactly when $\mathrm{Gal}(E/K) \trianglelefteq G$.

Part II: The Engine and the Proof

Dedekind: Independence of Characters

Lemma (Dedekind). Distinct field embeddings (*characters*) $\sigma_1, \dots, \sigma_n : E \to L$ are *linearly independent over* L: if $\sum_i a_i \sigma_i(x) = 0$ for all $x \in E^\times$ (with $a_i \in L$), then all $a_i = 0$.

Proof. Take a shortest nontrivial relation, $m \geq 2$ nonzero terms (one term would force $a_1 = 0$ as $\sigma_1(1) = 1$). Pick y with $\sigma_1(y) \neq \sigma_m(y)$. Replacing x by yx: $\sum_i a_i \sigma_i(y) \sigma_i(x) = 0$; multiplying the relation by $\sigma_m(y)$: $\sum_i a_i \sigma_m(y) \sigma_i(x) = 0$. Subtract:

$$\sum_{i=1}^{m-1} a_i \big(\sigma_i(y) - \sigma_m(y)\big) \sigma_i(x) = 0,$$

a shorter relation with nonzero leading coefficient $a_1(\sigma_1(y) - \sigma_m(y))$ — contradiction. $\square$

Artin's Theorem

Theorem (Artin). Let $G = \{\sigma_1 = \mathrm{id}, \ldots, \sigma_n\}$ be a *finite group of automorphisms* of a field E, and $K = E^G$ its fixed field. Then

$$[E : K] = |G| = n, \qquad E/K \text{ is Galois}, \qquad \mathrm{Gal}(E/K) = G.$$

A finite automorphism group recovers its fixed field exactly — the engine that powers the correspondence. Proof in two bounds: first $[E : K] \leq n$ (independence of characters, which also gives finiteness), then $[E : K] \geq n$ (degree bound).

Artin: $[E:K] \le |G|$

Show any $n+1$ elements $u_1, \ldots, u_{n+1} \in E$ are K-dependent. The $n \times (n+1)$ system $\sum_j \sigma_i(u_j)\, c_j = 0$ $(i = 1, \ldots, n)$ has a nonzero solution $(c_j) \in E^{n+1}$ (more unknowns than equations). Take one with the **fewest** nonzero entries; scale $c_1 = 1$.

If some $c_j \notin K$, say $\sigma(c_2) \neq c_2$ for $\sigma \in G$: applying σ permutes the equations, so $(\sigma(c_j))$ solves the same system; then $(c_j - \sigma(c_j))$ is a *shorter* nonzero solution (the $j = 1$ term vanishes, the $j = 2$ term does not) — contradiction.

So all $c_j \in K$, and the $\sigma_1 = \mathrm{id}$ equation $\sum_j u_j c_j = 0$ is a nontrivial K-dependence. Hence $[E:K] \le n$ — in particular $[E:K]$ is finite.

Artin: $[E : K] \geq |G|$, and Conclusion

Now that $[E : K]$ is finite: every σ_i fixes $K = E^G$, so $G \subseteq \mathrm{Aut}(E/K)$, and the degree bound gives

$$n = |G| \leq |\mathrm{Aut}(E/K)| \leq [E : K].$$

Combined with $[E : K] \leq n$: $[E : K] = n$, and the chain forces $|\mathrm{Aut}(E/K)| = [E : K]$, so E/K is *Galois*. Since $G \subseteq \mathrm{Aut}(E/K)$ with $|G| = n = |\mathrm{Aut}(E/K)|$, we get $G = \mathrm{Gal}(E/K)$. $\square$

Proof of the Correspondence

$\mathrm{Gal}(E/E^H) = H$**.** H is a finite automorphism group with fixed field E^H; Artin's theorem gives $\mathrm{Gal}(E/E^H) = H$.

$E^{\mathrm{Gal}(E/K)} = K$**.** First E/K is Galois (E/F normal $\Rightarrow$ E/K normal; E/F separable $\Rightarrow$ E/K separable). Apply Artin to $H' = \mathrm{Gal}(E/K)$:
$[E : E^{H'}] = |H'| = |\mathrm{Gal}(E/K)| = [E : K]$. With $K \subseteq E^{H'} \subseteq E$, this forces $E^{H'} = K$.

So the maps are inverse bijections. **Degrees:** $[E : K] = |\mathrm{Gal}(E/K)| = |H|$, and $[K : F] = [E : F]/[E : K] = |G|/|H| = [G : H]$. $\square$

Lattice Operations and a Finiteness Corollary

The bijection respects the lattice operations. For intermediate K_1, K_2 with $H_i = \mathrm{Gal}(E/K_i)$:

$$\mathrm{Gal}(E/K_1K_2) = H_1 \cap H_2, \qquad \mathrm{Gal}(E/K_1 \cap K_2) = \langle H_1, H_2 \rangle.$$

- σ fixes the compositum K_1K_2 (smallest field containing both) $\iff$ it fixes K_1 and K_2 $\iff$ $\sigma \in H_1 \cap H_2$.
- $E^{\langle H_1, H_2 \rangle} = E^{H_1} \cap E^{H_2} = K_1 \cap K_2$, so $K_1 \cap K_2 \leftrightarrow \langle H_1, H_2 \rangle$.

So the correspondence is a **lattice anti-isomorphism**: compositum $\leftrightarrow$ intersection, intersection $\leftrightarrow$ join.

Corollary. A finite group has finitely many subgroups, so a finite Galois extension has only *finitely many intermediate fields*.

Normal Subgroups $\leftrightarrow$ Normal Subextensions

Theorem. For finite Galois E/F, $G = \mathrm{Gal}(E/F)$, and $K \leftrightarrow H = \mathrm{Gal}(E/K)$:

$$K/F \text{ normal (equivalently Galois)} \iff H \trianglelefteq G,$$

and then restriction $\sigma \mapsto \sigma|_K$ gives an isomorphism

$$\mathrm{Gal}(K/F) \cong G/H = G/\mathrm{Gal}(E/K).$$

(Separability of K/F is automatic from that of E/F, so here "normal" and "Galois" coincide.)

Normal $\leftrightarrow$ Normal: Proof

Embeddings of K are restrictions of G. Each F-embedding $K \to \overline{F}$ extends along E/K (isomorphism extension theorem) to an embedding $E \to \overline{F}$, which lies in G since E/F is normal. So they are exactly $\{\sigma|_K : \sigma \in G\}$.

The equivalence. K/F normal $\iff \sigma(K) = K$ for all $\sigma \in G$ (normality = every F-embedding has image K). By the conjugation identity $\mathrm{Gal}(E/\sigma K) = \sigma H \sigma^{-1}$ and the bijection, $\sigma(K) = K \iff \sigma H \sigma^{-1} = H$. Hence "$\sigma(K) = K\ \forall \sigma$" $\iff H \trianglelefteq G$.

The quotient. For $H \trianglelefteq G$, restriction $\sigma \mapsto \sigma|_K$ maps $G \to \mathrm{Gal}(K/F)$, onto (every τ extends to G), with kernel $\{\sigma : \sigma|_K = \mathrm{id}\} = \mathrm{Gal}(E/K) = H$. So $\mathrm{Gal}(K/F) \cong G/H$. $\square$

Part III: Reading the Lattice

Abelian: $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$ (V_4)

$G = \mathrm{Gal}(E/\mathbb{Q}) \cong V_4 = \{\mathrm{id}, a, b, c\}$ ($a : \sqrt{3} \mapsto -\sqrt{3}$; $b : \sqrt{2} \mapsto -\sqrt{2}$; $c = ab$).
Subfield lattice (subgroups top$\to$bottom: $\{1\}$; $\langle a \rangle, \langle b \rangle, \langle c \rangle$; G — inclusion-reversed):

$$\begin{tikzcd}
 & E \arrow[dl, dash] \arrow[d, dash] \arrow[dr, dash] & \\
\mathbb{Q}(\sqrt{2}) \arrow[dr, dash] & \mathbb{Q}(\sqrt{3}) \arrow[d, dash] & \mathbb{Q}(\sqrt{6}) \arrow[dl, dash] \\
 & \mathbb{Q} &
\end{tikzcd}$$

G abelian $\Rightarrow$ every subgroup normal $\Rightarrow$ every intermediate field is Galois over $\mathbb{Q}$ (each degree 2).

Cyclic: $\mathbb{F}_{p^n}/\mathbb{F}_p$ — Subfields $\leftrightarrow$ Divisors

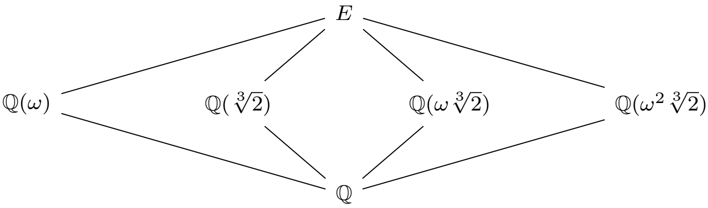
$\mathbb{F}_{p^n}/\mathbb{F}_p$ is Galois of degree n (the splitting field of $x^{p^n} - x$), with $G = \langle \varphi \rangle \cong \mathbb{Z}/n$ generated by the Frobenius $\varphi(x) = x^p$ (order n). For $d \mid n$, the subgroup $\langle \varphi^d \rangle$ (order n/d, index d) has fixed field $\{x : x^{p^d} = x\} = \mathbb{F}_{p^d}$ (degree d). So **subfields** $\leftrightarrow$ **divisors of** n, $\mathbb{F}_{p^d} \subseteq \mathbb{F}_{p^e} \iff d \mid e$.

$$
\begin{tikzcd}
 & \mathbb{F}_{p^6} \arrow[dl, dash] \arrow[dr, dash] & \\
\mathbb{F}_{p^2} \arrow[dr, dash] & & \mathbb{F}_{p^3} \arrow[dl, dash] \\
 & \mathbb{F}_p &
\end{tikzcd}
$$

The case $n = 6$: the subfield lattice is the divisor lattice of 6. (G cyclic, so every subextension is normal.)

Non-abelian: $\mathbb{Q}(\sqrt[3]{2}, \omega)/\mathbb{Q}$ (S_3)

$E = \mathbb{Q}(\sqrt[3]{2}, \omega)$, $[E : \mathbb{Q}] = 6$, $G \cong S_3$ permuting $r_1 = \sqrt[3]{2}, r_2 = \omega\sqrt[3]{2}, r_3 = \omega^2\sqrt[3]{2}$ $(\omega = e^{2\pi i/3})$. Subfield lattice:



Subgroups (top→bottom): $\{1\}$; $A_3 = \langle\rho\rangle$ and the three transposition subgroups $\langle\tau_i\rangle$; S_3. Degrees: $\mathbb{Q}(\omega)$ deg 2 ($\leftrightarrow A_3$, index 2); the three cubic fields deg 3 ($\leftrightarrow$ the order-2 subgroups).

S_3: Normal vs Non-normal

$A_3 \trianglelefteq S_3$ (index 2) $\Rightarrow \mathbb{Q}(\omega)/\mathbb{Q}$ is normal/Galois, degree 2, with $\mathrm{Gal}(\mathbb{Q}(\omega)/\mathbb{Q}) \cong S_3/A_3 \cong \mathbb{Z}/2$ ($\mathbb{Q}(\omega)$ splits $x^2 + x + 1$).

The three order-2 subgroups are not normal $\Rightarrow \mathbb{Q}(\sqrt[3]{2}),\ \mathbb{Q}(\omega\sqrt[3]{2}),\ \mathbb{Q}(\omega^2\sqrt[3]{2})$ are *not* normal over $\mathbb{Q}$ — the failure seen directly via $x^3 - 2$ (a root, but not all roots). G permutes the three, matching the three conjugate subgroups.

The group's internal structure — which subgroups are normal, the conjugacy of the transpositions — *is* the field structure: which subfields are normal, the conjugacy of $\mathbb{Q}(\sqrt[3]{2})$ with its siblings — the platform for the applications to come: solvability by radicals and the insolvability of the quintic.